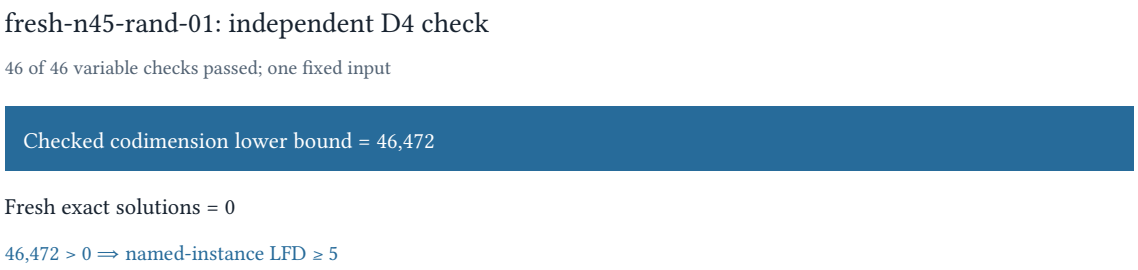
Certified n45 random01: LFD at least five

The independent D4 checker accepted all 46 variable multiplication checks for fresh-n45-rand-01. It certified codimension at least **46,472** and fresh exhaustive enumeration found **0** solutions. Thus the strict inequality establishes **LFD at least 5** for this named Boolean presentation.

The certificate is 1,045,099,360 bytes with SHA-256 bbc1bab615279e83f76d197d803f0ab2d8a41af7a90d566c472c23f80041b106. The collector checked all 47 worker artifacts and successful terminal return; three controls passed. Public certificate and evidence readback were verified.



Admission passed after 516.176 seconds at 47,017,943,040 bytes available. Producer and checker took 2078.219 and 509.419 seconds. The original panel now has 29 terminal, 22 certified LFD-at-least-five, and three remaining inputs. Frozen source 608be655740605d96d97157006c29894332a72db; protocol 9188cc34d411158bb01bd5c14387dd1f7c53bd610cec882c8780ad242b1e3b16.

Input n=45, k=23, seed 11859439544555254794, degree 4. Runtime command and paths are in report.md and the frozen plan. CPU time was not recorded.